

STEI, Romania

WMO: 151600

Lat: 46.5281N

Lon: 22.4664E

Elev: 279

StdP: 98.01

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.2	-8.0	-13.4	1.2	-8.9	-11.0	1.5	-6.3	8.2	1.3	7.2	1.5	1.6	180	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.3	32.2	20.8	30.2	20.6	28.5	19.9	22.2	28.6	21.5	27.9	20.7	27.0	2.4	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.2	15.4	24.7	19.4	14.7	24.0	18.6	13.9	23.3	66.4	28.7	63.7	27.8	61.1	27.3	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.9	5.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-0.3	3.0	6.2	10.8	15.7	19.3	20.4	20.9	15.9	11.2	7.1	1.8
	DBStd	8.25	4.74	4.40	4.27	4.74	3.20	3.23	2.88	3.37	4.20	4.40	4.65	4.23
	HDD10.0	1104	318	198	131	48	3	0	0	0	2	40	111	254
	HDD18.3	2919	576	429	376	227	92	27	13	13	97	221	337	512
	CDD10.0	1491	1	2	14	72	179	279	323	339	179	78	24	1
	CDD18.3	264	0	0	0	1	9	57	78	94	24	1	0	0
	CDH23.3	2664	0	0	2	31	127	521	779	948	232	24	0	0
	CDH26.7	858	0	0	0	1	12	141	255	383	67	1	0	0
Wind	WSAvg	2.3	2.2	2.4	2.3	2.3	2.3	2.1	2.2	2.1	2.1	2.3	2.4	2.3
Precipitation	PrecAvg	972	68	55	65	78	91	112	108	74	88	78	75	81
	PrecMax	1281	123	141	202	133	173	214	207	181	224	179	142	173
	PrecMin	696	29	11	20	21	38	35	48	10	14	7	5	7
	PrecStd	156	24	31	44	31	36	40	47	40	48	44	30	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	16.0	21.9	26.1	28.2	31.8	33.5	34.8	31.9	25.6	20.4	14.1
		MCWB	8.1	9.9	13.9	15.8	17.8	22.0	21.0	20.9	19.5	15.9	12.9	9.0
	2%	DB	9.3	13.5	18.2	23.9	26.4	30.0	31.3	32.7	28.1	23.7	18.1	11.1
		MCWB	6.5	8.6	10.8	14.4	17.7	21.5	20.5	20.5	18.4	15.5	12.3	7.7
	5%	DB	7.6	11.3	15.4	21.6	24.8	28.2	29.4	30.5	26.1	21.3	16.0	9.2
		MCWB	5.5	7.6	9.6	13.4	17.1	20.5	20.3	20.5	17.8	14.8	11.2	6.5
	10%	DB	6.0	9.4	12.9	19.1	23.0	26.6	27.6	28.4	23.3	18.4	14.0	7.5
		MCWB	4.4	6.5	8.2	12.1	16.3	19.8	19.6	20.2	17.0	13.6	10.5	5.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	10.9	13.7	16.6	19.9	23.7	22.7	23.1	20.4	17.3	13.7	10.0
		MCDB	11.6	14.6	20.5	24.2	25.3	29.5	28.9	30.6	29.4	22.3	18.2	13.2
	2%	WB	7.3	9.0	11.7	15.2	18.7	22.2	21.8	22.1	19.4	16.1	12.8	8.3
		MCDB	8.9	12.2	17.0	22.6	24.4	28.0	28.2	29.2	27.1	22.0	17.0	10.9
	5%	WB	5.7	8.1	10.1	14.0	17.7	21.3	21.2	21.2	18.4	15.2	11.9	6.8
		MCDB	7.3	11.1	14.7	20.3	23.1	27.0	27.4	28.3	24.5	20.2	15.4	8.9
	10%	WB	4.3	6.9	8.8	12.9	16.8	20.3	20.4	20.5	17.5	14.0	10.8	5.7
		MCDB	5.7	9.0	12.2	18.2	21.7	25.6	26.3	27.3	22.7	18.2	13.9	7.5
Mean Daily Temperature Range	5% DB	MDBR	6.0	7.6	9.7	10.9	11.1	11.5	12.2	12.3	10.7	10.4	7.6	5.5
		MCDBR	6.9	10.3	13.7	14.6	14.5	13.9	15.0	15.5	14.7	14.3	10.4	8.1
		MCWBR	4.5	6.1	7.4	6.7	7.0	6.3	6.2	5.8	6.0	7.2	5.6	5.1
	5% WB	MCDBR	6.4	8.8	12.4	13.2	12.7	12.9	13.1	13.7	12.9	12.8	9.3	7.6
		MCWBR	4.4	5.5	6.9	6.3	6.6	6.2	6.0	5.9	5.6	6.6	5.0	5.1
Clear-Sky Solar Irradiance	taub		0.326	0.356	0.401	0.456	0.441	0.436	0.453	0.458	0.421	0.383	0.353	0.327
	taud		2.308	2.241	2.175	2.052	2.161	2.215	2.174	2.157	2.224	2.318	2.346	2.346
	Ebn at Noon		749	798	813	798	829	834	812	789	783	759	711	704
	Edh at Noon		78	102	126	156	145	138	142	139	119	94	76	68
All-Sky Solar Radiation	RadAvg		1.20	2.00	3.15	4.35	5.25	5.80	5.67	5.24	3.72	2.55	1.46	0.94
	RadStd		0.17	0.26	0.46	0.45	0.50	0.48	0.50	0.50	0.42	0.30	0.20	0.12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)	+0.70	N/A	N/A	+0.99	N/A	N/A	-130	-197	+115	+58

Nomenclature: See separate page